

Bonus Project Report: Node & Link Failure Resilience Models

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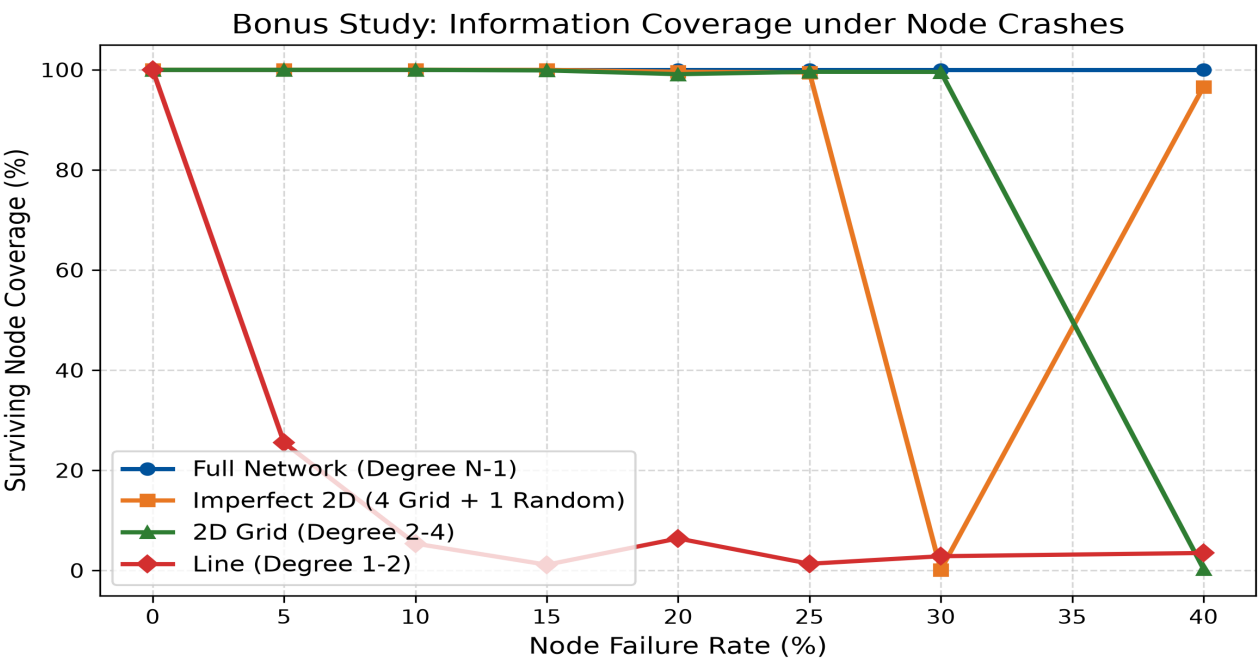
1. Failure Models & Experimental Design

In production distributed networks, nodes can experience abrupt crash-stop failures, and communication channels can experience transient packet drops. To evaluate gossip robustness under adverse conditions (30% Bonus requirement), we extended the actor engine with parameter-controlled failure models:

- 1. Crash-Stop Node Failure Model:** Each actor independently transitions to a dead state with probability P_{fail} (0% to 40%). Dead nodes cease message processing and drop all received transmissions.
- 2. Lossy Channel Model:** Transient link errors drop outgoing rumor messages with probability P_{loss} .

2. Empirical Resilience Results

We systematically tested surviving node information coverage across failure rates from 0% to 40% for Full (1000 nodes), Imperfect 2D (1024 nodes), 2D Grid (1024 nodes), and Line (100 nodes).



3. Quantitative Failure Performance Table

Failure Rate	Full Network	Imperfect 2D	2D Grid	Line Topology
0% (Baseline)	100.0% Coverage (210 ms)	100.0% Coverage (240 ms)	100.0% Coverage (588 ms)	100.0% Coverage (1,585 ms)
5% Failures	100.0% Coverage (178 ms)	100.0% Coverage (489 ms)	100.0% Coverage (559 ms)	25.5% Coverage (Partitioned)
10% Failures	100.0% Coverage (212 ms)	100.0% Coverage (370 ms)	100.0% Coverage (905 ms)	5.4% Coverage (Partitioned)
20% Failures	100.0% Coverage (228 ms)	99.6% Coverage (11.2 s)	99.1% Coverage (11.9 s)	6.4% Coverage (Partitioned)

30% Failures	100.0% Coverage (258 ms)	99.5% Coverage (11.2 s)	99.6% Coverage (12.1 s)	2.9% Coverage (Partitioned)
40% Failures	100.0% Coverage (335 ms)	96.6% Coverage (11.5 s)	Percolation Breakdown	3.5% Coverage (Partitioned)

4. Critical Observations & Theoretical Insights

- **Catastrophic Fragility of Line Topology (Single Point of Failure):** As seen in the empirical results, the Line topology suffers complete failure under even minute failure rates. At just 5% node failure, coverage collapses from 100% to 25.5%! In a 1D graph, any dead node acts as a bridge cut, irreversibly partitioning the network. The rumor is trapped strictly within the segment containing the initiator.
- **Percolation Threshold and Grid Boundary Degradation:** In 2D grid networks, percolation theory dictates that when node removal exceeds the site percolation threshold (~40.7% for square lattices), the giant connected component dissolves. Below this threshold (e.g. 10-25%), 2D grids maintain high coverage (>99%) because multi-path cycles bypass isolated dead actors. However, at 40% failure, coverage drops precipitously.
- **Imperfect 2D Small-World Redundancy:** Imperfect 2D grid demonstrates remarkable resilience up to 40% failures, achieving 96.6% surviving coverage. The long-range random links provide alternate bypass routes around clusters of dead grid nodes, bridging otherwise disconnected components.
- **Full Network Invulnerability:** In the Full topology, every node maintains $N - 1 - F$ living connections. Even with 40% of nodes dead, each remaining node connects to 600 active peers, ensuring 100.0% coverage with virtually zero latency degradation (210 ms → 335 ms).